Zoey Drassinower

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Zoey Drassinower (they/she) is a recent university graduate with B.Sc. in Computing, Math and Analytics. Completed multiple research contracts in graduate-level labs conducting novel research. Seeking to leverage strong analytical and communications skills in professional environments.

For technical articles, interactive demos, and other samples of my work, see my portfolio site ***www.zoeyd.ca***

Education

Bachelors in Computing (Honours) w/ Specialization in Computing and Mathematics from Queen's University

Academic Work Experience

Research Contracts

May 2023 – May 2024 **Summer Research/Undergraduate Final Project (High Dimensional Feature Selection),** ***Queen's University***

Applying statistical methods to identify key molecules in fungi and hot sauce.
Supervised by Prof. Randy Ellis, funded by NSERC USRA grant.

- Reviewed current literature to select algorithms from relevant academic papers
- Implemented and applied algorithms to very high dimensional data
- Visualized and presented results to peers and supervisors

May 2022 – Aug 2022 **Summer Research (Soccer Analytics),** ***Queen's University***

Modelling successful offensive progressions in soccer using player tracking data.
Supervised by Prof. Catherine Pfaff.

- Improved transformation algorithms, resulting in 20x faster code execution
- Migrated workflow to run in Docker containers

Teaching Assistant Contracts

Winter 2024 ***(TA)*** **Artificial Intelligence (300-level),** ***Queen's University***

Introduction to AI topics, such as Bayesian reasoning, automated planning and neural networks.
Course taught by Prof. Christian Muise.

- Standard TA responsibilities (grading, office hours, etc.)
- Identified and corrected bugs in auto-grader scripts
- Updated assignment code to be compatible with modern python versions

Fall 2023 ***(Head TA)*** **Programming Paradigms (300-level)** ***Queen's University***

Introduction to functional and declarative programming languages (Haskell, Prolog).
Course taught by Prof. Jana Dunfield.

- Standard TA responsibilities (grading, office hours, etc.)
- Allocated and managed responsibilities of other TAs
- Wrote and recorded videos teaching course materials

Fall 2022 Winter 2023 ***(TA)*** **Programming Paradigms (300-level)** ***Queen's University***

Introduction to functional and declarative programming languages (Haskell, Prolog).
Course taught by Prof. Jana Dunfield.

- Standard TA responsibilities (grading, office hours, etc.)
- Taught exam prep tutorials, created review resources

Teaching and Mentoring Experience

Jan 2024 – Apr 2024 **Introductory Programming Lessons**

- Taught weekly 1-on-1 programming lessons (in person) to a 1st year math student with no prior coding experience
- Designed a personalized lesson plan, tailored to meet the goals of the student
- Supplemented lessons with "workplace computing skills" such as debugging, documentation, style, and Git

Jun 2019 – Apr 2024 **Various Math and Physics Tutoring**

- Tutored students taking 1st and 2nd year math and physics courses

Grants and Awards

May 2023 **Undergraduate Summer Research Award** *(Natural Sciences and Engineering Research Council of Canada)* - **$9800**

- Awarded each year to ~3000 Canadian STEM undergraduates conducting novel research over the summer

Jan 2024 **Dean's Award of Excellence** *(Queen's University)*

- Awarded to undergraduates with GPA in the top 0.5% of GPAs in the Faculty of Arts and Sciences at Queen's

Independent Projects

For technical articles, interactive demos, and other samples of my work, see my portfolio site ***www.zoeyd.ca***

Py2048-ML v2 (Python)

- Re-creating the 2048 agent described in the Stochastic MuZero paper by Deep Mind
- Applies multiple state of the art techniques in deep reinforcement learning, such as reward scaling, prioritized replay buffers, Monte-Carlo tree search, residual networks, layer-normalization
- Neural networks built using TensorFlow

Zoelver (Python/Rust)

- Prototyped in Python and built in Rust – the Zoelver project is a framework/sandbox for exploring different algorithms in classical automated planning.
- Implements a variety of classical planning strategies, including Iterated Width and H-Add Heuristic Search.

Baseball Tracking Dashboard (Python)

- A custom data dashboard for visually analyzing batter tendencies
- Built and hosted via Streamlit
- Created as part of the 2024 Women in Sports Data Hackathon

Other Interests

Songwriting and Performance

- 15 years of vocal training at ProVoce Studios and Free Your Voice coaching
- Participated in Artist Development/Songwriter workshops at Lawson Vocal Studios
- Played hired performances at on-campus events throughout university
- Self-produced and recorded original song: Run Down

Competitive Ultimate Frisbee

- 1 season of varsity at Queen's University (2023-2024)
- 1 season of competitive tournaments w/ Toronto Remix (2024)
- Recently began volunteer coaching youth skills programs at Toronto Ultimate Club